MARKING RUBRIC

Machine_Learning_Assignment_01

Course: CSC354 - Machine Learning

Assessment-based marking guide with evaluation criteria

CLO-3 Apply unsupervised learning techniques to solve clustering problems.

Q1. Implement a K-means clustering algorithm in Python to cluster the Iris dataset.

Evaluation Criteria:

- **Algorithm/Approach (3 marks):**
  Correct algorithm or problem-solving approach
- **Code Correctness (3 marks):**
  Code produces correct output and handles edge cases
- **Logic & Efficiency (3 marks):**
  Logical flow and reasonable time/space complexity
- **Code Quality (2 marks):**
  Readable code with proper naming and structure

Expected Solution:
2

Total: 10 marks

Q2. Explain the concept of 'elbow method' in determining the optimal number of clusters for K-means algorithm and provide a Python code snippet to demonstrate it.

Evaluation Criteria:

- **Algorithm/Approach (3 marks):**
  Correct algorithm or problem-solving approach
- **Code Correctness (3 marks):**
  Code produces correct output and handles edge cases
- **Logic & Efficiency (3 marks):**
  Logical flow and reasonable time/space complexity
- **Code Quality (2 marks):**
  Readable code with proper naming and structure

Expected Solution:
2

Total: 10 marks

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By: Hadia Anwar